# Inspector's Brief: Demo District (Surveillance Zone A)

Date: October 2026 | Surveillance: Sentinel-2 RGB (10 m/pixel) | Model: ChimneyCensus-OBB v1.0 (Rotated IoU mAP: 0.834) | Officer: Inspector Farida Khan

KEY SURVEILLANCE FINDINGS:
- 65 unique active kilns detected across 120 tiles (~154 sq km), deduplicated across 30 px tile overlap boundaries.
- 18 (15–21) baseline 2023–24 FCBKs appear converted to cleaner Zigzag designs (mandate-compliant).
- 27 persistent illegal FCBKs remain in operation post the 2022 ban; 8 circular kilns (CFCBK) require heritage review.
- 51 kilns are clustered within 1 km across 18 local groups, allowing Farida to inspect multiple sites per vehicle patrol.

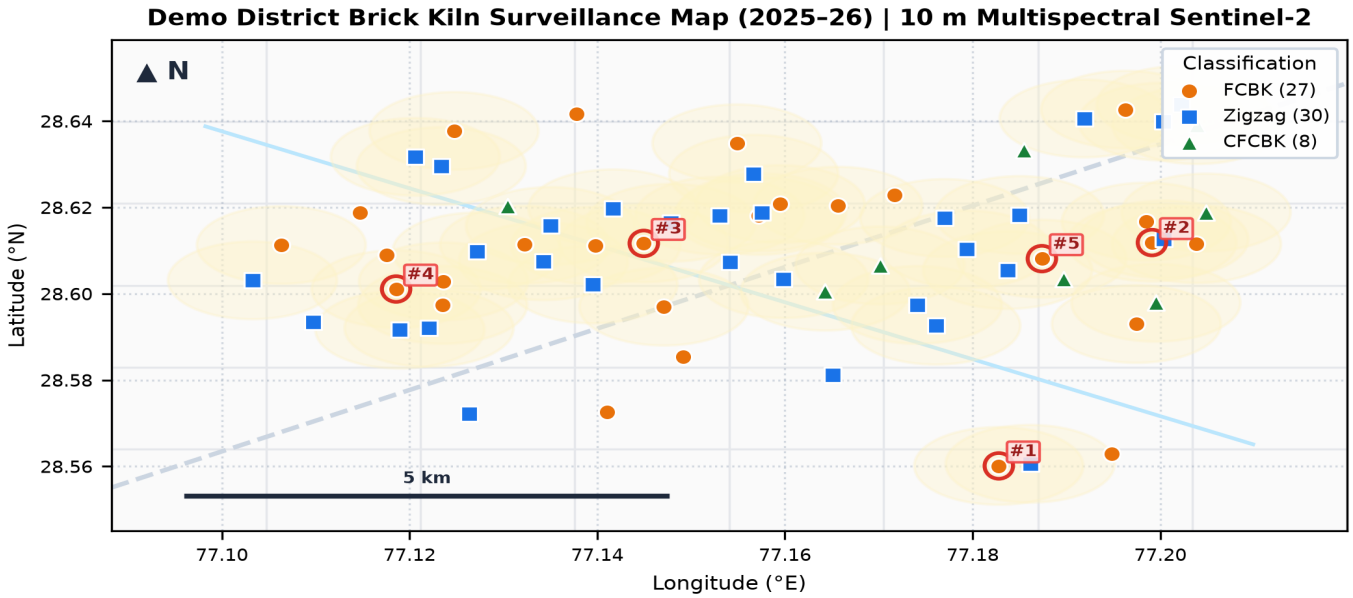



**Table A: Kiln Counts by Design Type & Honest Uncertainty Bounds**

| Type | Detected | Est. True Count (95% CI) | Precision | Recall | Note |
|---|---|---|---|---|---|
| FCBK | 27 | 18 – 41 | 0.89 | 0.85 | Banned 2022 design |
| Zigzag | 30 | 21 – 45 | 0.65 | 0.62 | Mandate compliant |
| CFCBK | 8 | 3 – 14 | 0.90 | 0.95 | Circular kiln (~3%) |
| **Total** | **65** | **42 – 100** | **-** | **-** | **All types combined** |

*Method: Estimated true count adjusts raw detections for held-out validation precision and recall, using 1,000 bootstrap resamples for the 95% interval.*

**Table B: Conversion Status of Historical (2023–24) FCBK Kilns**

| Matched 2023–24 FCBKs | Now Detected as Zigzag (Low–High) | Still FCBK | Not Found / Unclear |
|---|---|---|---|
| 45 (35 matched) | 18 (15–21) | 15 (Violation) | 12 |

*Conversion represents a 2023–24 registered FCBK site matched within ~30 m that now presents rectangular Zigzag geometry. Range reflects matching uncertainty.*

# Field Inspection Dispatch Plan & Top Priority Targets

**Table C: Priority Proximity Clusters (Within 1 km of Neighbor) — Total: 51 kilns in 18 clusters. Full registry in brief/outputs/within_1km.csv.**

| Cluster | Kilns | Types | Centre lat/lon | Max gap in m |
|---|---|---|---|---|
| Cluster 1 | 5 | 3 FCBK, 1 Zigzag, 1 CFCBK | 28.61431, 77.20135 | 958 m |
| Cluster 2 | 5 | 3 Zigzag, 1 CFCBK, 1 FCBK | 28.64119, 77.19892 | 1,187 m |
| Cluster 3 | 4 | 2 FCBK, 2 Zigzag | 28.61151, 77.13852 | 1,140 m |
| Cluster 4 | 4 | 3 FCBK, 1 Zigzag | 28.61951, 77.15996 | 866 m |
| Cluster 5 | 3 | 3 FCBK | 28.60046, 77.12191 | 634 m |

**Table D: Top 20 Kilns Farida Should Visit First (Actionable Dispatch Order)**

| # | Kiln ID | Type | Lat (°N) | Lon (°E) | Conf | Why visit |
|---|---|---|---|---|---|---|
| 1 | KILN-DEMO-025 | FCBK | 28.56011 | 77.18279 | 0.96 | Still FCBK post-2022 ban; high conf (0.96); near 1 kiln |
| 2 | KILN-DEMO-053 | FCBK | 28.61186 | 77.19911 | 0.95 | Still FCBK post-2022 ban; high conf (0.95); near 4 kilns |
| 3 | KILN-DEMO-048 | FCBK | 28.61167 | 77.14496 | 0.90 | Still FCBK post-2022 ban; high conf (0.90); near 3 kilns |
| 4 | KILN-DEMO-056 | FCBK | 28.60109 | 77.11857 | 0.91 | Still FCBK post-2022 ban; high conf (0.91); near 3 kilns |
| 5 | KILN-DEMO-052 | FCBK | 28.60813 | 77.18738 | 0.87 | Still FCBK post-2022 ban; high conf (0.87); near 3 kilns |
| 6 | KILN-DEMO-030 | FCBK | 28.61137 | 77.13226 | 0.90 | Still FCBK post-2022 ban; high conf (0.90); near 4 kilns |
| 7 | KILN-DEMO-032 | FCBK | 28.62078 | 77.15950 | 0.93 | Still FCBK post-2022 ban; high conf (0.93); near 5 kilns |
| 8 | KILN-DEMO-023 | FCBK | 28.61803 | 77.15719 | 0.85 | Still FCBK post-2022 ban; high conf (0.85); near 5 kilns |
| 9 | KILN-DEMO-028 | FCBK | 28.63773 | 77.12474 | 0.87 | Still FCBK post-2022 ban; high conf (0.87); near 2 kilns |
| 10 | KILN-DEMO-047 | FCBK | 28.61114 | 77.13983 | 0.85 | Still FCBK post-2022 ban (conf 0.85); near 6 kilns |
| 11 | KILN-DEMO-042 | FCBK | 28.61674 | 77.19852 | 0.84 | Still FCBK post-2022 ban (conf 0.84); near 4 kilns |
| 12 | KILN-DEMO-057 | FCBK | 28.60287 | 77.12362 | 0.81 | Still FCBK post-2022 ban (conf 0.81); near 4 kilns |
| 13 | KILN-DEMO-027 | FCBK | 28.59299 | 77.19744 | 0.81 | Still FCBK post-2022 ban (conf 0.81); near 1 kiln |
| 14 | KILN-DEMO-044 | FCBK | 28.61129 | 77.10634 | 0.56 | Still FCBK post-2022 ban (conf 0.56); near 1 kiln |
| 15 | KILN-DEMO-024 | FCBK | 28.59741 | 77.12353 | 0.71 | Still FCBK post-2022 ban (conf 0.71); near 4 kilns |
| 16 | KILN-DEMO-039 | FCBK | 28.62041 | 77.16564 | 0.79 | Still FCBK post-2022 ban (conf 0.79); near 4 kilns |
| 17 | KILN-DEMO-021 | FCBK | 28.64266 | 77.19630 | 0.80 | Still FCBK post-2022 ban (conf 0.80); near 4 kilns |
| 18 | KILN-DEMO-059 | FCBK | 28.59700 | 77.14710 | 0.71 | Still FCBK post-2022 ban (conf 0.71); near 1 kiln |
| 19 | KILN-DEMO-020 | FCBK | 28.63486 | 77.15489 | 0.72 | Still FCBK post-2022 ban (conf 0.72); near 1 kiln |
| 20 | KILN-DEMO-054 | FCBK | 28.61150 | 77.20386 | 0.38 | Unverified FCBK; low conf (0.38) check; near 4 kilns |

*Ranking score = Violation type (+40 FCBK, +35 CFCBK, +10 Zigzag) + Proximity (+20 within 1 km) + High confidence (+25 >= 0.85) or Verification urgency (+15 < 0.40).*

HOW SURE ARE WE? (LIMITATIONS & FIELD GUIDANCE):

• 10 m Resolution: Sentinel-2 pixels span 10×10 m. Subtle internal draft retrofits inside oval walls may look like FCBKs; field checks resolve this.

• 2025–26 Domain Shift: Seasonal soil and atmospheric variations may cause minor classification drift compared to training imagery.

• No Ground Truth for Demo District: Estimates rely on held-out regional validation evidence; borderline cases (< 0.50 conf) require ground confirmation.